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Predicting Compressive Strength of Concrete Incorporating Fly Ash, Blast Furnace Slag, and Superplasticizer Using Machine Learning Techniques

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Abstract

Concrete is the second most essential element in the construction industry, and its strength requirements vary based on the specific conditions of each project. However, determining the compressive strength of concrete involves laboratory tests, which wastes a lot of time and money. Researchers have developed machine learning models that predict the compressive strength of cement-based concrete having various mixes. In this research, the compressive strength of concrete incorporating fly ash, blast furnace slag, and superplasticizer is predicted using different machine learning models, namely, Linear Regression, Random Forest Regression, Decision Tree Regression, Extreme Gradient Boosting, Light Gradient Boosting, AdaBoost, and CatBoost Regression. Then the model's performance is evaluated based on prediction accuracy and prediction error rates i.e. R square, MSE, RMSE, MAE, MSLE, and RMSLE. Comparing all seven models, CatBoost and XGBoost have shown the highest prediction accuracy ($R^2 = 0.8948$, $MAE = 4.034$, $RMSE = 5.766$ and $R^2 = 0.890$, $MAE = 3.978$, $RMSE = 5.881$) respectively. In conclusion, the CatBoost and XGBoost are the best machine learning models for predicting the compressive strength of concrete with the highest prediction accuracy, and lowest modeling error.

Keywords: Compressive Strength, Concrete, Machine Learning, Prediction, CatBoost, XGBoost

1 INTRODUCTION

Concrete is one of the most versatile and widely used construction materials on earth. It is strong, durable[1], low maintenance, fire resistant, simple to use, and can be made to fit any shape and size, from the unfathomably massive to the humble steppingstone. However, none of those advantages would matter without this, it is cheap. Concrete has four main ingredients: water, fine aggregates, coarse aggregates, and cement. But in certain instances, according to the site conditions and materials availability, some other elements like fly ash, blast furnace slag, and superplasticizers are also added to enhance the mechanical properties of concrete[2].

Among the valuable engineering properties of concrete, there is an important property known as concrete compressive strength, that defined as the ability of concrete to withstand specific compressive forces, and this is a widely used parameter in structural engineering for the design of reinforced concrete structures. Compressive strength test in laboratories[3] is carried out by preparing small cubes of concrete, which are then tested using compression testing machines after 7, 14, and 28 days. The formula used is:

$$\text{Compressive Strength} = \frac{\text{Force}}{\text{Area}} \quad (1)$$

This physical test is simple but a resource-consuming process in terms of time and money. To deal with this, researchers have developed machine-learning techniques for predicting the compressive strength of concrete[4][5][6][7][8]. Many models and algorithms have been developed that predict the compressive strength of cement-based concrete rapidly and accurately. Machine learning is the science of programming machines to think and act like humans without being specifically programmed. It uses different algorithms to learn tasks, fed with data from which they learn to perform these tasks. Machine learning is a subset of artificial intelligence (AI), a science concerned with imparting human-like intelligence onto machines that can sense, reason, act, and predict.

2 OBJECTIVE

The objectives of this research are:

1. To develop an efficient, cost-effective, and precise model that can predict the compressive strength of the concrete incorporating fly ash, blast furnace slag, and superplasticizer with higher accuracy.
2. To develop a fast and reliable tool that streamlines the research process.

3 METHODOLOGY

The research was conducted using the following steps:

3.1 Data Collection

The data for this research is downloaded directly from the UC Irvine Machine Learning Repository[9]. This data is compiled by Professor I-Cheng Yeh and is freely available for any kind of research. This dataset contains a total of 1030 instances with 9 variables. The input variables are cement (kg/m^3), blast furnace slag (kg/m^3), fly ash (kg/m^3), water (kg/m^3), coarse aggregate (kg/m^3), fine aggregate (kg/m^3), and age (days), and the output variable being predicted compressive strength (MPa).

3.2 Programming Workflow and Tools Used

Python was chosen for this research because of its extensive libraries and frameworks. It is highly versatile and widely used in data science and machine learning.

All the models are built in Jupyter Notebook, as it is the best suited for Python. It allows to run code cells one at a time, add headings, and save models.

Python libraries that were utilized are:

1. Pandas: For handling the data.
2. Numpy: For performing numerical operations.
3. Sikit-Learn: For building and testing Machine Learning Models.

3.3 Data Separation as X and Y

From the database, the input variables were considered (x) which are cement, blast furnace slag, fly ash, water, coarse aggregate, fine aggregate, and age. The output variable was considered (y) which is predicted compressive strength.

3.4 Data Splitting and Training

In this study, 90% of the data is regarded as the training data, while the remaining 10% is regarded as the testing data[10]. All the models were trained on the training data and then testing data were used to gauge the accuracy of the resulting model.

3.5 Model Building

The following models were employed to predict the compressive strength of concrete:

3.5.1 Linear Regression

It is a linear approach for modeling the relationship between a dependent and one or more independent variables[11].

3.5.2 Random Forest Regressor

It is a type of machine learning model that uses an ensemble of decision trees to make its prediction[12][13]. It is built by taking a random sample of data and then building an ongoing series of decision trees on the subsets and outputs the average prediction from all the trees to improve accuracy and control overfitting.

3.5.3 Decision Tree Regressor

It builds models in the form of a tree structure. It breaks down a dataset into smaller and smaller subsets while at the same time, an associated decision tree is incrementally developed. The result is a tree with decision nodes and leaf nodes[14]. A decision node has two or more branches, each representing values for the attribute tested. The leaf node represents a decision on the numerical target. The topmost decision node in a tree which corresponds to the best predictor is called the root node.

3.5.4 XGBoost (eXtreme Gradient Boosting)

It is an advanced implementation of a gradient-boosting algorithm. It repeatedly builds new models and combines them into an ensemble model[15]. It Initially builds the first model and calculates the error for each observation in the dataset. Then it builds a new model to predict those residuals. And finally, it adds prediction from this model to the ensemble model.

3.5.5 Light Gradient Boost Regression

It is also a boosting algorithm. Its decision tree grows leaf-wise, based on the condition only one leaf splits[15].

3.5.6 Ada Boost Regressor

It is again a boosting algorithm. It is a meta-estimator that begins by fitting a regressor on the original dataset and then fits additional copies of the regressor on the same dataset but where the weights of instances are adjusted according to the error of the current prediction[15].

3.5.7 Cat Boost Regressor

It is a high-performance gradient-boosting technique. Its decision tree grows symmetric. It integrates the predictions from all the trees when making predictions, creating models that are extremely accurate and reliable[15].

3.6 Evaluating Model Performance

The following statistical tools[16] were utilized to check the accuracy of each model:

3.6.1 Coefficient of Determination (R-squared)

It indicates the proportion of variance in the dependent variable that is predictable from the independent variables [17]. The R-squared value ranges from 0 to 1. Its higher values suggest that the model is performing well.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

3.6.2 Mean Squared Error (MSE)

It is the average of the squared difference between predicted values and actual values. It is the average magnitude of the error. Higher values indicate less accuracy.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3)$$

3.6.3 Root Mean Squared Error (RMSE)

It is the square root of the mean squared error (MSE), in the same units as the target variable, sensitive to outliers. Higher values indicate less accuracy.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}} \quad (4)$$

3.6.4 Mean Absolute Error (MAE)

It is the average of absolute differences between predicted values and actual values. Higher values indicate less accuracy.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (5)$$

3.6.5 Mean Squared Log Error (MSLE)

It measures the averages of the squared differences between the logarithms of the predicted values and actual values. It penalizes underestimation more than overestimation. Higher values indicate less accuracy.

$$MSLE = \frac{1}{n} \sum_{i=1}^n (\log(1 + \hat{y}_i) - \log(1 + y_i))^2 \quad (6)$$

3.6.6 Root Mean Squared Log Error (RMSLE)

It is the square root of the mean squared logarithmic error (MSLE). It is useful for relative error measurement. Higher values indicate less accuracy.

$$RMSLE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\log(1 + \hat{y}_i) - \log(1 + y_i))^2} \quad (7)$$

Where $\hat{y}_i$ and y_i is the predicted and experimental value of the variable Compressive Strength, respectively, n is the number of data, and $\bar{y}$ is the average model output value representing the Compressive Strength and experimental value, respectively.

4 RESULTS

The results of this study are summarized in two tables, one for the training dataset and the other for the testing dataset. Each table presents the performance metrics for all seven models, including Mean Squared Error (MSE), Coefficient of determination (R-square), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Squared Logarithmic Error (MSLE), and Root Mean Squared Logarithmic Error (RMSLE).

4.1 Performance Metrics for Training Dataset

This table should include all the metrics for all the models based on their performance during the training phase.

Method	MSE	R2	RMSE	MAE	MSLE	RMSLE
Linear Regression	104.890855	0.617915	10.24162366	8.15419297	0.111453935	0.333847174
Random Forest	120.485276	0.561109	10.97657851	8.73900195	0.13278485	0.364396556
Decision Tree	141.869573	0.483213	11.91090144	9.46526796	0.151673376	0.389452662
XGBoost Regressor	13.4937217	0.950847	3.67338015	2.76582102	0.018821184	0.137190321
LGBoost Regressor	26.0804441	0.904997	5.106901617	3.93386565	0.027156146	0.164791219
AdaBoost Regressor	46.7530154	0.829693	6.837617667	5.71687541	0.074883877	0.273649186
CatBoost Regressor	14.9617114	0.945499	3.868037156	2.98304765	0.017381757	0.131839891

4.2 Performance Metrics for Testing Dataset

This table should include the same metrics for all the models but based on their performance during the testing phase.

Method	MSE	R2	RMSE	MAE	MSLE	RMSLE
Linear Regression	129.17391	0.591365	11.3654699	8.6610062	0.09766709	0.312517347
Random Forest	138.91084	0.560563	11.7860444	8.9765508	0.09919316	0.314949451
Decision Tree	162.14364	0.487067	12.7335635	9.664055	0.11662621	0.341505802
XGBoost Regressor	34.592638	0.890568	5.88155066	3.9786313	0.03165609	0.177921579
LGBoost Regressor	49.325527	0.843961	7.02321343	5.2342893	0.03929174	0.198221435
AdaBoost Regressor	61.153394	0.806545	7.82006352	6.352236	0.05939529	0.243711481
CatBoost Regressor	33.253442	0.894805	5.76657974	4.0341874	0.03059594	0.174916944

The performance trends reveal that CatBoost and XGBoost performed well and maintained their accuracy in the training and testing phases. This consistency in performance suggests that these models are less prone to overfitting and are more generalizable to unseen data. The superior performance of these models suggests they could be handy in real-world scenarios, reducing the need for time-consuming and costly laboratory tests. These models offer valuable insights for practical applications in civil engineering, where precise estimation of compressive strength is crucial.

5 CONCLUSION

Predicting the compressive strength of concrete rapidly and accurately has great significance in the engineering field, especially for researchers. This study focused on developing seven machine-learning models, namely, Linear Regression, Random Forest, Decision Tree, XGBoost, LightGBM, AdaBoost, and CatBoost Regressor to predict the compressive strength of concrete, using a dataset of 1030 instances. The data was divided into 90% for training and 10% for testing[10] to ensure thorough model training and evaluation of unseen data.

Among all the seven models, four were gradient boosting models[18] (XGBoost, LightGBM, AdaBoost, and CatBoost) that achieved strong predictive accuracy, with R-squared values exceeding 0.8, and low RMSE and MAE values, which reflect precise predictions with minimal error.

Two models, in particular XGBoost and CatBoost, emerged as the best performers, delivering almost identical results with exceptionally high R-squared values and low RMSE and MAE values. For the training data set, the R-square values are 0.945 and 0.950, MAE values 2.983 and 2.765, and RMSE values are 3.868 MPa and 3.673 MPa, respectively for CatBoost and XGBoost, which means the test's predicted values and original values are comparable. For the testing data, CatBoost and XGBoost models achieved almost the same R-square values of 0.894 and 0.890, MAE values of 4.031 MPa and 3.978 MPa, and RMSE values of 5.766 MPa and 5.881 MPa respectively. These top two models demonstrated superior accuracy in predicting compressive strength.

The overall success of the boosting models, especially XGBoost and CatBoost two top-performing models, underscores their potential for practical applications in the engineering field. This study highlights the importance of careful model selection and fine-tuning in developing reliable predictive tools for the accurate estimation of concrete compressive strength.

6 RECOMMENDATIONS

In this study, the models were trained using a dataset of 1030 compressive strength values, ranging from 2.3318 MPa to 82.5992 MPa. To improve the accuracy of the models, it is recommended to increase the size and diversity of the dataset.

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